

Chapter 14

Multiple integrals

Example 1

$$\int_0^1 xy^2 dx$$

$$= y^2 \int_0^1 x dx$$

$$= y^2 \left. \frac{x^2}{2} \right|_0^1$$

$$= y^2 \left(\frac{(1)^2}{2} - \frac{(0)^2}{2} \right)$$

$$= \frac{y^2}{2}$$

$$\int_0^1 xy^2 dy$$

$$= x \int_0^1 y^2 dy$$

$$= x \left. \frac{y^3}{3} \right|_0^1$$

$$= x \left(\frac{(1)^3}{3} - \frac{0}{3} \right)$$

$$= \frac{x}{3}$$

Example 2

$$(a) \int_1^3 \int_2^4 (40 - 2xy) dy dx$$

Soln:-

$$= \int_1^3 \int_2^4 (40 dy) dx - \int_1^3 \int_2^4 (2xy dy) dx$$

$$= \int_1^3 (40x) dx - \int_1^3 \left(2x \frac{y^2}{2} \right) dx$$

$$= \int_1^3 (160 - 80) dx - \int_1^3 (16x - 4x) dx$$

$$= \int_1^3 80 dx - \int_1^3 12x dx$$

$$= 80x \Big|_1^3 - \left(\frac{12x^2}{2} \right) \Big|_1^3$$

$$= (240 - 80) - (54 - 6)$$

$$= 160 - 48$$

$$= 108$$

or

$$\int_2^4 \int_1^3 (40 - 2xy) dx dy$$

Soln -

$$= \int_2^4 \int_1^3 (40 dx) dy - \int_2^4 \int_1^3 (2xy dx) dy$$

$$= \int_2^4 (40x) \Big|_1^3 dy - \int_2^4 \left(\frac{2x^2 y}{2} \right) \Big|_1^3 dy$$

$$= \int_2^4 (120 - 40) dy - \int_2^4 (9y - y) dy$$

$$= \int_2^4 80 \, dy - \int_2^4 8y \, dy$$

$$= 80y \Big|_2^4 - 4y^2 \Big|_2^4$$

$$= (320 - 160) - (64 - 32)$$

$$= 160 - 32$$

$$= 128$$

Example 3

Use double integral to find volume of the solid that is bounded above by the plane $z = 4 - x - y$ & below by the rectangle $R = [0, 1] \times [0, 2]$

$$\int_0^1 \int_0^2 (4 - x - y) \, dy \, dx$$

$$\text{or} \int_0^2 \int_0^1 (4 - x - y) \, dx \, dy$$

Sol:-

$$= \int_0^2 \int_0^2 (4-x-y) dx dy$$

$$= \int_0^2 \int_0^2 (4 dx) dy - \int_0^2 \int_0^2 x dx dy - \int_0^2 \int_0^2 y dx dy$$

$$= \int_0^2 4x \Big|_0^2 dy - \int_0^2 \frac{x^2}{2} \Big|_0^2 dy - \int_0^2 yx \Big|_0^2 dy$$

$$= \int_0^2 4 dy - \int_0^2 \frac{1}{2} dy - \int_0^2 y dy$$

$$= (4y \Big|_0^2) - \left(\frac{1}{2} y \Big|_0^2 \right) - \left(\frac{y^2}{2} \Big|_0^2 \right)$$

$$= 8 - 1 - 2$$

$$= \boxed{5} \text{ Ans}$$

Example 4

Evaluate double integral

$$\iint xy^2 dA$$

over the rectangle

$$R = \{ (x, y) : -3 \leq x \leq 2, 0 \leq y \leq 1 \}$$

$$\int_{-3}^2 \int_0^1 xy^2 dy dx$$

$$\int_{-3}^2 \left[\frac{xy^3}{3} \right]_0^1 dx = \int_{-3}^2 \frac{x}{3} dx$$

$$\left[\frac{x^2}{6} \right]_{-3}^2 = \frac{4}{6} - \frac{9}{6} = -\frac{5}{6}$$

$$0 -$$

$$\int_{-3}^2 \int_0^1 xy^2 dy dx$$

Ex 14.1 double integral properties
 Example 1-4 Q1-11, Q16

$$= \int_3^2 \left(\frac{xy^3}{3} \Big|_0^1 dy \right) dx$$

$$= \int_3^2 \frac{x}{3} dx$$

$$= \left[\frac{x^2}{6} \right]_3^2$$

$$= \frac{4}{6} - \frac{9}{6}$$

$$= -\frac{5}{6}$$

14.1 Q1) - 612) Evaluate the iterated integrals.

$$1) \int_0^1 \int_0^2 (x+3) dy dx$$

Sol:

$$= \int_0^1 \left(\int_0^2 (x+3) dy \right) dx$$

$$= \int_0^1 (x+3) dy \Big|_0^2$$

$$= (x+3) \int_0^2 dy$$

$$= \int_0^1 x dx$$

$$= (x+3) \cdot y \Big|_0^2$$

$$= \int_0^1 \left[\int_0^2 (x+3) dy \right] dx$$

$$= \int_0^1 \left. x+3(y) \right|_0^2 dx$$

$$= \int_0^1 (x+3)(2) dx$$

$$= \int_0^1 (2x+6) dx$$

$$= \int_0^1 2x dx + \int_0^1 6 dx$$

$$= \left\{ x^2 \right|_0^1 + 6x \right|_0^1$$

$$= (1^2 - 0^2) + 6(1) - 6(0)$$

$$= 1 + 6$$

$$= 7$$

$$2) \int_1^3 \int_{-1}^1 (2x-4y) dy dx$$

Sol:-

$$= \int_1^3 \left[\int_{-1}^1 (2x-4y) dy \right] dx$$

$$= \int_1^3 \left[2xy \right|_{-1}^1 - 2y^2 \right|_{-1}^1 dx$$

$$= \int_1^3 [2x(1) - (-1)] - 2(1^2 - (-1)^2) dx$$

$$= \int_1^3 (2x(2) - 2(0)) dx$$

$$= \int_1^3 4x \, dx$$

$$= 2x^2 \Big|_1^3$$

$$= 2(3)^2 - (1)^2$$

$$= 2(8)$$

$$= 16 \text{ Ans.}$$

$$3) \int_2^4 \int_0^1 x^2 y \, dx \, dy$$

Sol:-

$$= \int_2^4 \left[\int_0^1 x^2 y \, dx \right] dy$$

$$= \int_2^4 \left[\frac{x^3 y}{3} \Big|_0^1 \right] dy$$

$$= \int_2^4 \frac{y}{3} [(1)^3 - (0)^3] dy$$

$$= \int_2^4 \frac{y}{3} dy$$

$$= \frac{y^2}{6} \Big|_2^4$$

$$= \frac{1}{6} [(4)^2 - (2)^2]$$

$$= \frac{12}{6}$$

$$= 2 \text{ Ans}$$

$$4) \int_{-2}^0 \int_{-1}^2 (x^2 + y^2) dx dy$$

Sol:-

$$= \int_{-2}^0 \left[\int_{-1}^2 (x^2 + y^2) dx \right] dy$$

$$= \int_{-2}^0 \left[\int_{-1}^2 x^2 dx + \int_{-1}^2 y^2 dx \right] dy$$

$$= \int_{-2}^0 \left[\frac{(2)^3 - (-1)^3}{3} + y^2 [(2) - (-1)] \right] dy$$

$$= \int_{-2}^0 \left[\frac{9}{3} + 3y^2 \right] dy$$

$$= \int_{-2}^0 (3 + 3y^2) dy$$

$$= \left[3y + y^3 \right]_{-2}^0$$

$$= 3(0 + 2) + (0)^3 - (-2)^3$$

$$= +6 + 8$$

$$= 14 \text{ Ans.}$$

$$⑤ \int_0^{\ln 3} \int_0^{\ln 2} e^{x+y} dy dx$$

Sol:

$$= \int_0^{\ln 3} \left[\int_0^{\ln 2} e^{x+y} dy \right] dx$$

$$= \int_0^{\ln 3} e^x \cdot (e^{\ln 2} - e^0) dy$$

$$= \int_0^{\ln 3} e^x (2-1) dy$$

$$= \int_0^{\ln 3} e^x dy$$

$$= e^x \Big|_0^{\ln 3}$$

$$= e^{\ln 3} - e^0 = 3 - 1 = 2$$

$$\textcircled{6} \int_0^2 \int_0^1 y \sin x dy dx$$

Sol:

$$= \int_0^2 \left[\int_0^1 y \sin x dy \right] dx$$

$$= \int_0^2 \left[\frac{y^2}{2} \cdot \sin x \right]_0^1 dx$$

$$= \int_0^2 \frac{\sin x \cdot (1)^2 - (0)^2}{2} dx$$

$$= \int_0^2 \frac{\sin x}{2} dx$$

$$= \frac{1}{2} \int_0^2 \sin x dx$$

$$= \frac{1}{2} (-\cos x) \Big|_0^2 = \frac{1}{2} [-\cos(2) + \cos(0)]$$

$$= \frac{1 - \cos 2}{2} \text{ Ans.}$$

$$7) \int_{-1}^0 \int_2^5 dx dy$$

Sol:-

$$= \int_{-1}^0 x \Big|_2^5 dy$$

$$= \int_{-1}^0 (5-2) dy$$

$$= \int_{-1}^0 3 dy$$

$$= 3y \Big|_{-1}^0 \Rightarrow = 3(0) - (-1) = 3 \text{ Ans.}$$

$$8) \int_4^6 \int_{-3}^7 dy dx$$

$$= \int_4^6 \left[\int_{-3}^7 dy \right] dx \Rightarrow = \int_4^6 y \Big|_{-3}^7 dx \Rightarrow = \int_4^6 (7 - (-3)) dx$$

$$= \int_4^6 10 dx \Rightarrow 10x \Big|_4^6 \Rightarrow 10(6-4) \Rightarrow = 20$$

$$9) \int_0^1 \int_0^1 \frac{x}{(xy+1)^2} dy dx$$

$$= \int_0^1 \left[x \int_0^1 \frac{1}{(xy+1)^2} dy \right] dx$$

$$= \int_0^{\pi/2} x \cos x \, dx \Rightarrow u = x, \, du = dx, \, v = \sin x$$

$$\int x \cos x \, dx \Rightarrow x \sin x \Big|_{\pi/2}^{\pi} - \int_{\pi/2}^{\pi} \sin x \, dx$$

Sol:-

$$= \int_0^1 \int_0^1 (xy+1)^{-2} x \, dy \, dx$$

$$= \int_0^1 \left[\frac{(xy+1)^{-2+1}}{-2+1} \right]_0^1 dx \Rightarrow \int_0^1 \left[-1(xy+1)^{-1} \right]_0^1 dx$$

$$= \int_0^1 \left[\frac{-1}{xy+1} \right]_0^1 dx$$

$$= \int_0^1 \left[\frac{-1}{x(1)+1} - \frac{-1}{x(0)+1} \right] dx$$

$$= \int_0^1 \left(\frac{-1}{x+1} + 1 \right) dx$$

$$= \int_0^1 \left(\frac{1}{x+1} - 1 \right) dx$$

$$= \left[\ln(x+1) - x \right]_0^1$$

$$= \left[(\ln(1+1) - 1) - (\ln(0+1) - 0) \right]$$

$$= \left[\ln(2) - 1 - 0 \right]$$

$$= 1 - \ln 2$$

Note
for Q11 $= \int e^{f(x)} f'(x) dx$
 $= e^{f(x)}$

$$7) 10) \int_{\pi/2}^{\pi} \int_1^2 x \cos xy \, dy \, dx$$

Sol:-

$$= \int_{\pi/2}^{\pi} x \left[\frac{\sin xy}{x} \right]_1^2 dx \Rightarrow \int_{\pi/2}^{\pi} (\sin 2x - \sin x) dx$$

$$= \left[-\frac{\cos 2x}{2} + \cos x \right]_{\pi/2}^{\pi} = \left(-\frac{\cos 2\pi}{2} + \cos \pi \right) - \left(-\frac{\cos \pi}{2} + \cos \frac{\pi}{2} \right)$$

$$= \left(-\frac{(1)}{2} - 1 \right) - \left(-\frac{(-1)}{2} + 0 \right)$$

$$= -\frac{3}{2}$$

$$11) \int_0^{\ln 2} \int_0^1 x y e^{xy^2} \, dy \, dx$$

$$= \int_0^{\ln 2} \int_0^1 e^{xy^2} x y \, dy \, dx \Rightarrow \int_0^{\ln 2} \frac{1}{2} \int_0^1 e^{xy^2} 2xy \, dy \, dx$$

$$= \int_0^{\ln 2} \frac{1}{2} \left[e^{xy^2} \right]_0^1 dx \Rightarrow \int_0^{\ln 2} \frac{1}{2} (e^{x(1)^2} - e^{x(0)}) dx$$

$$= \int_0^{\ln 2} \frac{1}{2} (e^x - 1) dx \Rightarrow \frac{1}{2} [e^x - x]_0^{\ln 2}$$

$$= \frac{1}{2} [e^{\ln 2} - \ln 2] - \frac{1}{2} [e^0 - 0] = \frac{1}{2} [2 - \ln 2 - 1]$$

$$= \frac{1 - \ln 2}{2}$$

14.1 Q16

Evaluate the double integral over the rectangular region R .

$$\iint_R (x \sin y - y \sin x) dA$$

$$R = \{x, y : 0 \leq x \leq \pi/2, 0 \leq y \leq \pi/3\}$$

Solⁿ:-

$$= \int_0^{\pi/2} \int_0^{\pi/3} (x \sin y - y \sin x) dy dx$$

$$= \int_0^{\pi/2} \left[-x \cos y - \sin x \frac{y^2}{2} \right]_0^{\pi/3} dx$$

$$= \int_0^{\pi/2} \left(-x \cos \frac{\pi}{3} - \frac{\sin x}{2} \left(\frac{\pi}{3} \right)^2 \right) - \left[-x \cos(0) - \frac{\sin x}{2} (0)^2 \right]$$

$$= \int_0^{\pi/2} \left(-\frac{x}{2} - \frac{\pi^2}{18} \sin x + x \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{x}{2} - \frac{\pi^2}{18} \sin x \right) dx$$

$$= \left[\frac{x^2}{4} - \frac{\pi^2}{18} (-\cos x) \right]_0^{\pi/2}$$

$$= \left[\frac{(\pi/2)^2}{4} + \frac{\pi^2}{18} \cos \left(\frac{\pi}{2} \right) \right] - \left[\frac{(0)^2}{4} + \frac{\pi^2}{18} \cos(0) \right]$$

$$= \frac{\pi^2}{16} + 0 - 0 - \frac{\pi^2}{18}$$

$$= \frac{9\pi^2 - 8\pi^2}{144} = \frac{\pi^2}{144} \text{ Ans.}$$